Abstract

This study presents a comparative, longitudinal analysis of visual content published across ten Telegram channels (five Russian, five Ukrainian) during the Russo-Ukrainian conflict from February 2022 to February 2026. Drawing on a corpus of approximately 24,000 scraped media items, it applies a structured automated annotation framework across seven dimensions: content type, implied violence, gamification, dehumanization, aestheticization, narrative framing, and camera frame. Despite operating in sharp ideological opposition, Russian and Ukrainian channels share a common visual vocabulary, differing not in the types of imagery deployed but in their narrative function. Russian channels progressively oriented toward political commentary and ideological framing, while Ukrainian channels anchored in occupied territories maintained a documentary register defined by destruction and civilian harm. Explicit violent imagery is present in 41% of all posts, yet Ukrainian channels score marginally higher on implied violence, reflecting the material conditions of occupied communities rather than propagandistic intent. The rise of drone POV footage to 12% of the corpus by 2024 produced the sharpest qualitative shifts observed, with gamification and dehumanization scores significantly elevated relative to all other frame types. Regression analysis identifies dehumanization as the central mediating variable across the annotation network, with implications for media literacy and counter-disinformation policy.

Introduction

Since Russia's full-scale invasion of Ukraine on 24 February 2022, Telegram has functioned as the primary arena of visual narrative contestation in the conflict. With top pro-Kremlin channels accumulating 95.5 billion views in 2022 alone and a 481% surge in Telegram citations across pro-Russian media (DFRLab, 2024), the platform has become not merely a conduit for wartime communication but a structural component of the conflict itself. Yet despite a growing body of scholarship on digital propaganda, disinformation, and platform-mediated witnessing, no study has systematically examined the visual content of this war at scale, across both sides, and over its full duration.

This study addresses that gap. Drawing on a corpus of approximately 24,000 media items scraped from ten Telegram channels (five Russian, five Ukrainian) spanning February 2022 to February 2026, it applies a structured automated annotation framework across seven analytical dimensions: content type, implied violence, gamification, dehumanization, aestheticization, narrative framing, and camera frame. By coding these dimensions symmetrically across Russian and Ukrainian channels, ranging from state-affiliated institutions to independent milbloggers, the study offers a comparative and longitudinal account of how wartime visual strategies have evolved and diverged across four years of conflict.

The research is motivated by three central questions. First, how do Russian and Ukrainian Telegram channels differ in their visual content strategies, and have those differences intensified or converged over time? Second, how has the technical mode of visual production shifted, particularly with the rise of drone warfare? Third, what structural relationships exist between the dimensions of visual propaganda, and what do those relationships reveal about how meaning is constructed in wartime imagery? The findings contribute to scholarship on conflict media, platform studies, and visual communication, while demonstrating the

methodological feasibility of large-scale automated image analysis as a tool for conflict research.

Literature Review

The Russo-Ukrainian conflict is frequently characterised as a "hybrid war" in which digital platforms serve as primary theatres of narrative control, with Telegram emerging as its most consequential visual arena (Bareikytė & Skop, 2022). A foundational precedent was established by Makhortykh and Sydorova (2017), who demonstrated that image-based communication during the 2014 Donbas conflict intensified polarisation rather than producing shared understanding. The scale of Telegram's subsequent growth is documented by DFRLab (2024), whose analysis reveals a 481% increase in Telegram citations across pro-Kremlin media, with top channels accumulating 95.5 billion views in 2022 alone; RIA Novosti nearly tripled its annual viewership by 2023, while anonymous milblogger channels leveraged their opacity as a trust-building branding tactic. This infrastructure of mass visual dissemination was not improvised: Theisen et al. (2025) found an 8,925% surge in posts and a 5,352% increase in images shared by Russian milblogger accounts in the two weeks preceding the February 2022 invasion, suggesting that coordinated visual escalation on Telegram functions as a structural precursor to conflict rather than merely its by-product.

Research into the content of wartime visual production reveals deeply asymmetric framing strategies on each side. Ptaszek et al. (2024) found that Russian institutional coverage maintained a stable technical-military register insulated from battlefield realities, while Ukrainian output remained more event-responsive — demonstrating how Telegram amplifies propagandistic framing rather than correcting for it. At the grassroots level, Bareikytė and Makhortykh (2024) show that the platform simultaneously enables authentic civilian testimony and strategic propaganda within overlapping channel ecosystems, a duality they term "inconspicuous digital witnessing." Visually, content on both sides relies on metonymic strategies — ruins and soldiers substituting for abstract concepts of destruction — producing imagery that is "agonised, localized, and politicized" and designed to bridge psychological distance for international audiences (Nagy-Béni, 2025). Official and independent accounts alike deploy "memetic warfare," mobilising archetypal hero/victim/villain narratives to shape opinion, with victim-centred imagery shown to generate significantly higher engagement (Mejova et al., 2025).

The broader visual economy of the conflict is constrained by a crisis of representation and severe epistemological instability. Conventional coverage defaults to sensationalist clichés — tanks, explosions, devastation — that flatten the complexity of lived conflict (Sopova, 2021), while "war influencers" must navigate what Heřmanová et al. (2025) term "ambivalent visibility," reconciling the ethics of witnessing with a platform logic that rewards spectacular subject matter. This tension is compounded by a disinformation environment of unprecedented scale: DFRLab (2024) documents how anonymous channels use their opacity to circulate manipulated imagery that is structurally difficult to attribute or refute, and Standaert (2024) shows that Telegram functions as an "anteroom" where fabricated content consolidates before migrating to mainstream platforms. The deployment of deep fakes — including those targeting President Zelenskyy — further demands computational detection methods to distinguish authentic from synthetic footage (Boháček & Farid, 2022), while the platform's archival instability makes civic archiving essential for preserving wartime visual evidence (Bareikytė & Skop, 2022).

The existing scholarship, while establishing critical foundations, leaves a significant methodological gap. Studies are predominantly qualitative, focus on a single actor or platform, or rely on text-mining that treats visual content as secondary. No study has systematically annotated a large-scale, longitudinal corpus of visual media drawn simultaneously from Russian and Ukrainian Telegram channels across the full duration of the 2022–2026 conflict. This research addresses that gap by scraping and annotating approximately 24,000 media items from ten channels — five Russian, five Ukrainian — spanning state-affiliated institutions to independent milbloggers, and coding for violence, aesthetic register, and content type over time. By applying this framework symmetrically to both sides, the study offers a comparative, longitudinal account of how visual strategies have evolved and mutually responded across the war's duration — directly engaging questions raised but unanswered by prior work, including the DFRLab's (2024) identification of viewership trends and channel volatility, by grounding aggregate platform patterns in granular visual content analysis.

Methodology

Data Collection

This study draws on a corpus of media posts scraped from ten Telegram channels affiliated with Russian military actors and war correspondents active during the conflict in Ukraine. The channels were selected to represent a range of pro-Russian and pro-Ukrainian information ecosystems, including front-line reporting accounts, official military-linked channels, and occupied-territory outlets. Data collection was conducted using a custom Python script built on the Telethon library, which interfaces with the Telegram API to retrieve message metadata programmatically.

The scraping pipeline was designed to collect media-bearing posts: photographs, videos - by retrieving their thumbnail, and documents, published between 24 February 2022 and 24 February 2026. For each message, the following metadata was extracted: channel identifier, publication date, media type, file size, view count, forward count, reply count, and user reactions. To ensure temporal representativeness while managing data volume, a systematic sampling method was applied, selecting up to 50 evenly spaced media posts per channel per month, resulting in a corpus of approximately 24 000 sampled observations.

Content Annotation

A structured annotation framework was developed to systematically code each media item across seven analytical dimensions, enabling a standardised comparative analysis of visual content across channels and over time. The codebook was designed to capture both descriptive and interpretive features of each image or video. The seven dimensions are as follows: (1) content type, identifying the primary subject of the media (e.g. combatants, equipment, territory, casualties); (2) implied violence, assessing the degree to which the content suggests or depicts harm; (3) gamification, coding for visual or rhetorical elements that frame warfare in ludic or competitive terms; (4) dehumanization, identifying representations that strip enemy actors of human attributes; (5) aestheticization, measuring the degree to which violence or destruction is rendered visually appealing or heroic; (6) narrative framing, capturing the broader discursive register of the post (e.g. victorious, defensive, martyrological); and (7) camera frame and point of view, recording compositional and perspectival choices that shape viewer identification.

Annotation Pipeline

Annotation was conducted using a Python-based pipeline that submitted each image to a large language model via API, returning a JSON object with a coded value for each of the seven dimensions. To inform the categories of the seven dimensions, a first exploratory 2D map was developed, enabling a first overview of the data's clusters. The pipeline was developed in three stages. A first batch was annotated using the Claude API as a gold standard reference. The research team then manually coded 50 images, which were fed into the model as few-shot demonstrations to transfer codebook logic to the automated pipeline. Final annotation at scale was carried out using GPT-4.5. A recurring challenge was the model's reluctance to code violent content due to built-in safety constraints; this was addressed by prompting the model to adopt the persona of an academic researcher in conflict studies, which proved effective in eliciting consistent annotations across the full dataset.

2D map

The 2D visual map was produced by passing 400 images through CLIP (ViT-B-32), a vision-language model. The 400 images were manually selected to ensure representativity both across channels and across content type. The pipeline was implemented as a Python script executed in Google Colab, using the sentence-transformers library for CLIP encoding and scikit-learn for t-SNE.

Channel Content Analysis

The primary goal of this part of the analysis is to identify shifts in communication strategies across various Telegram channels. By normalizing the data (posts by channel and month) into percentages, the relative importance of different content types (e.g., "combat", "civilian victims", "propaganda") can be compared.

The data is rendered as interactive Plotly heatmaps with a shared colour scale (ranging from 0% to 80%). The colour scale is capped at 80%, which accommodates the extreme dominance of certain categories in certain channels (notably the political category in wargonzo, which reaches values above 60% in some months) while preserving gradient resolution for the more commonly observed range of 0–40%.

Each cell in the heatmaps represents the percentage share of a given content type within a given channel in a given month.

The interface offers two complementary views. The "category" view presents one heatmap per content type, with channels on the y-axis and the monthly timeline on the x-axis, allowing cross-channel comparison for a single category. The "channel" view presents one heatmap per channel, with content categories on the y-axis, allowing a holistic reading of each channel's editorial profile over time.

Violence Analysis

Of the seven annotated dimensions, implied violence was selected for dedicated in-depth analysis. While all dimensions contribute to the broader analytical framework, violence is the dimension most directly constitutive of wartime visual communication — present across content types, channels, and time periods in ways that cut across the other analytical categories. Its dedicated treatment is further justified by the density of interpretive information available: beyond the numerical score, each annotation includes a justification text describing the specific visual cues that informed the rating, enabling qualitative analysis at scale.

The analysis is structured in two stages. The first is quantitative, examining four aspects of the violence scores across the corpus: their overall distribution, their evolution over the four-year observation period from February 2022 to February 2026, their variation across channels and between Russian and Ukrainian channel groups, and their relationship to post engagement as measured by view counts. The second stage is qualitative, constructing an inductive typology of violence by systematically reading annotation descriptions and justification texts and grouping images by shared visual logic — rather than relying solely on the model's content type classifications. This process identified eight distinct visual genres through which violence is represented across the corpus, ranging from direct casualty imagery to aestheticised memorial memes.

All results are presented through an interactive web-based visualisation developed specifically for this analysis, allowing for dynamic exploration of the data across time, channel, and violence category.

Analysis of technical means when representing the war

To track shifts in the technical mode of visual representation over time, images were grouped into half-year periods (H1/H2) from the first half of 2022 through the second half of 2025, yielding eight time bins with roughly 2,300 to 3,500 images each. The partial 2026 period (989 images) was excluded to avoid truncation bias.

The primary dependent variable is frame type, a seven-category classification of how each image was technically produced: drone POV, soldier POV, civilian recording, professional production, composite/meme, screenshot, and other. For each period, proportional shares were calculated at both the aggregate corpus level and per individual channel to distinguish corpus-wide trends from channel-specific behavior. This two-level approach is necessary because aggregate shifts can be driven by a small number of high-volume channels rather than reflecting a uniform change across the information ecosystem.

As a secondary lens, mean scores for the five continuous annotation dimensions (implied violence, gamification, dehumanization, aestheticization, narrative framing) were computed per period and cross-tabulated by frame type. This allows assessment of whether changing frame composition also corresponds to qualitative shifts in how violence is represented — specifically, whether the rise of drone footage coincides with increased gamification and dehumanization.

Correlatory Effects

To explore correlatory effects across the five annotation dimensions, we conducted a series of bivariate ordinary least squares (OLS) regression analyses on the full annotated corpus of 25,474 images. For each model, we designated one score as the independent variable and a theoretically related score as the dependent variable, drawing on prior literature on visual propaganda and conflict media to motivate each pairing. We computed Pearson correlation coefficients, regression slopes, intercepts, and R^2 values, and assessed statistical significance via p-values derived from the t-distribution on the regression slope.

We constructed three targeted models. First, we regressed implied violence onto gamification score to test whether images framing warfare as ludic, through HUD overlays and targeting graphics, were systematically more violent. Second, we regressed implied violence onto dehumanization score to examine whether images depicting people as targets or anonymous

casualties, including "ГРУЗ 200" composites and bloodied faces, drove violence scores upward. Third, we regressed narrative framing onto aestheticization score to test whether visual polish and stylization functioned as a carrier signal for ideological messaging.

To contextualise each relationship, we computed mean scores per discrete score level and plotted 95% confidence intervals using $\pm 1.96 \times (\sigma/\sqrt{n})$. We disaggregated results by content type, frame type, and channel, visualising group means as bubble charts weighted by sample size. Finally, we produced a full 5×5 Pearson correlation heatmap to identify second-order relationships across the score network. Dehumanization consistently functioned as the central mediating variable, correlating with every other dimension at $r = 0.43$ to 0.59.

Findings

2D map

The map reveals two overlapping axes of clustering. The first is content-based: images group into broad visual categories regardless of source channel, including geographic maps and infographic cards, military portraiture, hardware and weapons systems, and scenes of physical destruction. The second axis, partly overlaid on the first, reflects channel affiliation. Russian channels cluster heavily around maps and operational infographics, heroic soldier portraiture, and technological warfare imagery. This visual repertoire displays the war as strategic and controlled, highlighting the territorial gains made and framing the soldier as a heroic agent, in line with Soviet propaganda conventions. Ukrainian channels, by contrast, cluster around portraits of dead Russian soldiers and scenes of physical destruction photographed in a documentary register, oriented toward witness and accountability rather than narrative triumph.

Channel Content Analysis

Initial observations suggest that in the early stages of the conflict (Q1/Q2 2022), content types were highly fragmented. Over time, channels began to “specialize” their messaging, indicated by consistent horizontal bands of colour in the channel-view heatmaps.

The most structurally significant finding is the divergence between Ukrainian and Russian channels in their balance of political versus kinetic content. Russian channels — particularly wargonzo and RVvoenkor — show a sustained and growing orientation toward political commentary, with wargonzo reaching 40–67% political content share for most of the observation period. The other Russian channels follow the same broad pattern at lower magnitudes, suggesting this is a feature of the Russian military blogger ecosystem as a whole rather than individual editorial choices.

Ukrainian channels display more variation. A_shtirlitz is the notable exception among them, combining high political shares with exceptional propaganda meme volumes in 2022 — a combination consistent with an explicit information warfare function. Other Ukrainian channels, such as kherson_non_fake and hyevuy_dnepr, are dominated by destruction and civilian content, reflecting their geographic anchoring to areas under sustained bombardment and their role as local documentation outlets.

One of the clearest temporal trends across all channels is the emergence of drone strike content from late 2022 onwards, growing steadily through 2023–2025. This directly tracks

the escalating centrality of drone warfare to the conflict and represents one of the few moments where the data reflects a material change in the war rather than editorial strategy.

The civilian category remains consistently more prominent across Ukrainian channels than Russian ones, suggesting that humanizing the civilian population is a structural Ukrainian narrative priority — one that Russian channels largely subordinate to military, political, and ideological content.

Taken together, the heatmaps reveal a set of differences in how Ukrainian and Russian Telegram channels have approached content in this conflict; and how each channel has its own "posting culture". Across both groups, the gradual rise of drone strike content tracks the material evolution of the conflict itself.

Violence Analysis

Across the full corpus of 26,707 annotated items, the mean implied violence score is 4.23 out of 10, with a median of 4. The distribution is strikingly bimodal: scores cluster around the 1–4 range and then spike again at 6 and 8, with score 5 forming a notable valley. This pattern likely reflects a threshold effect in the annotation model, which appears to treat moderate violence as categorically distinct from explicit violence rather than as a midpoint on a continuum. Most consequentially, 41% of all images score 6 or above — the threshold at which violence shifts from implied to explicit — meaning that nearly one in two posts across these channels depicts violence in unambiguous terms.

Over the four-year observation period, mean violence scores rise from approximately 3.8 in February 2022 to stabilise around 4.3–4.6 from mid-2022 onwards, with a notable peak in early 2023 coinciding with the Battle of Bakhmut. Rather than escalating progressively, the visual register of violence appears to reach a plateau relatively early in the conflict and then persist — suggesting that the normalisation of violent imagery is rapid and durable rather than gradual.

Perhaps the most counterintuitive finding concerns the Russian-Ukrainian channel comparison. Ukrainian channels score marginally but consistently higher on implied violence (mean 4.31) than Russian channels (mean 4.16). This is explained by the composition of the Ukrainian channel group: donbassrealii and kherson_non_fake — both anchored in Russian-occupied territories — record the highest mean violence scores in the entire corpus (4.73 and 4.59 respectively). These channels document violence inflicted upon their communities, producing a visual register of destruction and aftermath that systematically outscores the operational and propagandistic content dominant on Russian channels. Violence, in other words, is not primarily a Russian narrative choice in this corpus — it is a Ukrainian documentary reality.

The qualitative typology identifies eight distinct visual genres of violence, ranging from drone strike footage — characterised by a gamified HUD aesthetic that abstracts killing into a targeting interface — to Груз 200 memorial memes, a genre unique to this conflict in which portrait photographs of soldiers are overlaid with flame effects and death notation, producing aestheticised grief without graphic imagery. Notably, the relationship between violence and engagement does not follow expected patterns: mean view counts peak at score 4 and decline at higher scores, suggesting that extreme violence circulates in narrower networks rather than driving mass reach.

Analysis of technical means when representing the war

The most significant structural shift across the corpus is the rise and partial retreat of drone POV footage. Drone imagery more than tripled from 3.4% of all images in early 2022 to a peak of 12% in the first half of 2024, before declining to 6.1% by late 2025. This surge was not uniform. Battlefield-oriented channels like ZA_FROHT, voenacher, and rusich_army drove it, reaching 29 to 31% drone content at peak, while commentary channels like a_shtirlitz and donbassrealii never exceeded 5%. The drone shift is a property of combat-proximate channels, not the ecosystem as a whole.

This rise came at the direct expense of civilian recording, which fell from 41% to 28%. The war's dominant visual perspective shifted from the horizontal, ground-level viewpoint of a bystander to the vertical, overhead viewpoint of a drone operator. The camera moved from witnessing violence to being embedded in the system producing it. Soldier POV, despite GoPro proliferation, declined from 3.5% to 2% — the drone displaced rather than complemented the helmet cam.

Critically, drone footage is not just more common but qualitatively different. Gamification scores within drone POV reached 6.0 out of 10 by late 2025, versus a corpus average of 1.7. Dehumanization hit 5.0 versus 2.1. These are not marginal gaps. HUD overlays, targeting reticles, and top-down framing reduce human figures to points of impact, producing a visual register closer to a video game than to documentation. The technical mode of image production is not neutral — the same event filmed from the ground and from a drone generates fundamentally different representations of violence.

The partial retreat from the 2024 peak reflects both electronic warfare countermeasures reducing successful strike recordings and a possible novelty decay, as composite and meme content rose from 11% to 16% simultaneously. The visual economy of the war does not follow a linear trajectory toward technological mediation — it exhibits a rise-plateau-retreat pattern shaped by operational constraints and audience dynamics. Aggregate gamification and aestheticization scores also trended upward across all frame types, suggesting a broader normalization of stylized violence representation independent of any single technology.

Correlatory Effects

Across the three regression models, results confirmed that all five annotation dimensions are meaningfully interconnected, though the strength and nature of each relationship differs substantially. Together, the models reveal a layered structure to the visual rhetoric of pro-Russian war media, in which dehumanization functions as the central organising variable, aestheticization and narrative framing operate as a tightly coupled ideological unit, and gamification represents a narrower, content-specific phenomenon.

The weakest of the three models examined the relationship between gamification score and implied violence score. We found a statistically significant but modest positive association ($r = 0.327$, $R^2 = 0.107$, $\beta = +0.370$, $p < 0.0001$), indicating that gamification explains approximately 10.7% of variance in implied violence. Disaggregation by content type revealed that this relationship is almost entirely driven by drone strike footage, which recorded the highest mean gamification score of any content type at 6.88. Other content types scoring very highly on implied violence, including casualty imagery and explosion footage, scored near zero on gamification. This suggests the gamified register is a specialised visual

grammar confined to a particular technological framing of warfare rather than a pervasive feature of violent imagery more broadly.

The second model, regressing implied violence onto dehumanization scores, produced substantially stronger results ($r = 0.567$, $R^2 = 0.321$, $\beta = +0.735$, $p < 0.0001$). Dehumanization explained 32.1% of variance in implied violence, approximately three times the explanatory power of gamification. The mean implied violence score rose steadily from 2.28 at dehumanization score zero to 9.0 at dehumanization score ten, with no plateau effect observed at any intermediate level. Casualty imagery was the key driver, recording a mean dehumanization score of 4.59 and a mean implied violence score of 7.78, the highest of any content type on the violence dimension. Across the full correlation heatmap, dehumanization emerged as the single most connected variable in the score network, correlating with every other dimension at $r = 0.43$ to 0.59, suggesting it functions as a mediating pathway through which other visual registers partly produce their effects on implied violence.

The third model examined the relationship between aestheticization score and narrative framing score and produced the strongest association of the three ($r = 0.581$, $R^2 = 0.338$, $\beta = +0.687$, $p < 0.0001$). Visual polish and ideological messaging are tightly coupled in this content ecosystem. Propaganda memes were the decisive outlier, recording mean scores of 6.64 on aestheticization and 7.27 on narrative framing, values entirely detached from every other content type in the corpus. Notably, screenshots including tactical maps recorded a narrative framing mean of 4.29 despite a modest aestheticization score of 3.40, indicating that ideological messaging can also travel through information design without requiring high visual production values.

Discussion and implications

The findings of this study both confirm and complicate the existing literature on wartime visual communication, while producing several results that resist easy theoretical assimilation. Taken together, they suggest that the visual dimension of the Russo-Ukrainian conflict operates according to logics that are neither straightforwardly propagandistic nor uniformly escalatory, and that the analytical frameworks inherited from prior work require revision when applied at scale.

The first significant finding concerns the shared visual grammar identified by the 2D cluster map. Despite the stark ideological opposition between Russian and Ukrainian channels, content groups primarily by type (maps, hardware, portraiture, destruction) rather than by channel affiliation. This challenges the intuition that wartime visual production is fundamentally differentiated along national or political lines. What diverges is not the visual vocabulary but its narrative deployment: Russian channels frame the same visual repertoire within a register of strategic control and heroic agency, while Ukrainian channels orient comparable imagery toward documentation and accountability. This distinction maps closely onto what Bareikytė and Makhortykh (2024) term the duality between propaganda and witnessing, but the present study adds a structural dimension. Both functions draw on a common pool of visual tropes, suggesting the conflict has generated a shared iconographic language that both sides inhabit differently rather than separately.

The violence analysis produces perhaps the most counterintuitive result. Ukrainian channels record marginally but consistently higher implied violence scores than Russian ones, driven

specifically by channels anchored in occupied territories. This finding inverts the common assumption that violent imagery is primarily a feature of Russian information aggression. The data instead reveals a structural asymmetry: Russian channels subordinate violence to operational and ideological framing, while Ukrainian channels in Kherson and Donbas document destruction as a lived and ongoing reality. This is consequential for how the international community interprets the visual output of this conflict, as violence here is not principally a rhetorical choice but an empirical condition of certain communities. The finding also extends Makhortykh and Sydorova's (2017) foundational work on polarisation through image-based communication, demonstrating that asymmetry in violent visual output reflects asymmetry in material experience rather than in propagandistic intent.

The drone analysis offers the study's clearest evidence of technology reshaping representation. The rise of drone POV footage to 12% of all images by 2024, concentrated in battlefield-proximate channels, corresponds to the sharpest qualitative shifts in the corpus. Gamification scores within drone content reach 6.0 out of 10, against a corpus average of 1.7, and dehumanization scores hit 5.0 versus 2.1. These are not marginal differences. The drone camera does not merely document violence; it reframes it within a targeting interface that strips human figures of individuality and consequence. This extends Boháček and Farid's (2022) concern with synthetic imagery to a broader argument: the most transformative representational shift in this conflict may not be deepfake fabrication but the routinisation of a real visual grammar that produces analogous epistemological effects through legitimate footage.

The regression analysis identifies dehumanization as the central mediating variable across the annotation network, correlating with every other dimension at $r = 0.43$ to 0.59. This has direct implications for counter-disinformation and media literacy interventions. If dehumanization functions as a structural carrier for the other dimensions of propaganda (violence, aestheticization, narrative framing) then targeting it specifically, rather than addressing individual content types in isolation, may offer greater analytical and policy leverage. Finally, the finding that engagement peaks at moderate violence scores (score 4) and declines at extreme levels challenges platform-logic assumptions about virality. Extreme violence appears to circulate within narrower, self-selecting networks rather than driving mass reach. This is a pattern with significant implications for how researchers and policymakers model the spread of harmful wartime content online, and one that warrants dedicated follow-up with larger engagement datasets than the present study can support.

Limitations

Several limitations must be acknowledged. First, the sampling method — capped at 50 posts per channel per month — ensures temporal comparability but precludes any analysis of absolute engagement or publication volume. All cross-channel and cross-temporal comparisons are therefore proportional rather than absolute.

Second, the automated annotation pipeline rests on a training set of only 50 manually coded images, which introduces subjectivity into the scoring framework. The codebook dimensions — particularly implied violence, aestheticization, and dehumanization — involve inherently interpretive judgments that are relative to the dataset and difficult to replicate without access to the original training material. A technical limitation was also observed whereby the model displayed a tendency to avoid extreme scores, particularly at the upper end of scales, which may compress variance in the results. Future iterations should incorporate larger and more systematically validated training sets, drawing on established academic literature on

automated image analysis in conflict contexts; improved annotation quality would also benefit from dedicated funding.

Third, the corpus is limited to ten Telegram channels. This captures only a fraction of the Russian military information ecosystem, which comprises a dense network of channels that actively repost and amplify each other's content. Channels that are less visible, operate under pseudonyms, or have since been deleted are systematically excluded, meaning the dataset skews toward the most prominent and stable accounts. Moreover, restricting the analysis to Telegram leaves aside other platforms — notably VKontakte, Twitter/X, and YouTube — that play significant roles in wartime visual communication.

A further limitation concerns the treatment of video content, which constitutes a significant share of posts on military Telegram channels. For operational and computational reasons, the pipeline processes only the thumbnail image extracted from each video rather than the full audiovisual file. This introduces a substantial reduction in analytical depth: a single static frame cannot capture narrative progression, motion effects, editing rhythm, or the affective dimensions of sound — including music, explosions, voiceovers, and silence — all of which are central to how war videos construct meaning and emotional impact

This study is best understood as an exploratory contribution rather than a definitive empirical account. Its primary value lies in demonstrating the methodological feasibility of large-scale automated visual analysis in conflict contexts, and in generating hypotheses for more rigorously resourced future research.

Conclusion and further research

This study set out to examine how visual content – including violence, aesthetic register, and content type are represented across a longitudinal corpus of visual media drawn from ten Telegram channels — five Russian, five Ukrainian — spanning the full four years of the conflict from February 2022 to February 2026. By applying a systematic automated annotation framework symmetrically across both sides, it offers what prior scholarship has not: a comparative, longitudinal account of how wartime visual strategies have evolved and diverged at scale.

Several findings stand out. The 2D visual map reveals that content clusters primarily by type rather than channel affiliation, suggesting that the visual grammar of this war — maps, portraiture, hardware, destruction — is shared across sides even where its narrative function differs sharply. The channel content analysis shows that Russian channels have progressively oriented toward political commentary at the expense of kinetic content, while Ukrainian channels anchored in occupied territories maintain a documentary register dominated by destruction and civilian harm. The violence analysis confirms that explicit violent imagery is pervasive — present in 41% of all posts — but also that its distribution is uneven in analytically significant ways: Ukrainian channels document violence as a lived reality, Russian channels largely subordinate it to operational and ideological framing, and extreme violence does not drive engagement in the way platform logic might predict.

Taken together, these findings are best understood as generative rather than conclusive. The study demonstrates the methodological feasibility of large-scale automated visual analysis in conflict contexts, and maps the terrain sufficiently to identify where deeper investigation is

warranted. Several directions for further research follow directly from its limitations and findings.

Future work should expand the corpus substantially — both in terms of channel volume, to capture the broader and more volatile ecosystem of smaller milblogger accounts, and in terms of platform diversity, extending the analysis to VKontakte, Twitter/X, and YouTube where complementary visual strategies operate. The annotation framework would benefit from grounding in established theoretical literature on visual framing, propaganda aesthetics, and conflict photography, moving beyond inductive category construction toward hypotheses derived from prior research. The treatment of video content in particular remains an unresolved limitation: analysing full audiovisual material rather than static thumbnails would substantially deepen the analysis of how sound, editing rhythm, and motion construct meaning in wartime posts. Finally, each analytical thread opened here — the gamification of drone warfare, the Груз 200 memorial genre, the relationship between violence and engagement, the editorial divergence between Russian and Ukrainian channels — merits dedicated follow-up research with the methodological rigour that an exploratory study of this kind cannot provide. The primary contribution of this work is to have made those threads visible.

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